



RB005 Week 3 — English Worksheet (Beginner)

Topic: Core Mechanic — Pick Your Pattern · **Course:** Dream Game Studio · **Level:** Beginner · **Time:** about 30 minutes

This week you script your game's **core mechanic** — the thing the player does again and again — using a **pattern** from your toolbox.

1 · Predict 🧠

Read the code. Before you run it in Studio, circle or write your answer.

```
local coin = script.Parent
local debounce = false

coin.Touched:Connect(function(hit)
    local player = game.Players:GetPlayerFromCharacter(hit.Parent)
    if player and not debounce then
        debounce = true
        player.leaderstats.Coins.Value += 1
        coin:Destroy()
    end
end)
```

You touch the coin. Do your Coins go up? Circle one: yes · no

Why?


```
local door = workspace.GoldDoor

door.Touched:Connect(function(hit)
    local player = game.Players:GetPlayerFromCharacter(hit.Parent)
    if player and player.leaderstats.Coins.Value >= 10 then
        door.CanCollide = false
    end
end)
```

You have 4 coins and touch the door. Does it open? Circle one: yes · no

2 · Find the Difference 🦁

Each pair shows clean code first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — One touch should give **1** coin, not many.

```
-- clean
local debounce = false

coin.Touched:Connect(function(hit)
    local player = game.Players:GetPlayerFromCharacter(hit.Parent)
    if player and not debounce then
        debounce = true
        player.leaderstats.Coins.Value += 1
        coin:Destroy()
    end
end)
```

```
-- buggy
coin.Touched:Connect(function(hit)
    local player = game.Players:GetPlayerFromCharacter(hit.Parent)
    if player then
        player.leaderstats.Coins.Value += 1
    end
end)
```

What is missing? How many coins can one touch give in the buggy code?

Pair B — The countdown should tick once per second.

```
-- clean
while timeLeft > 0 do
    print(timeLeft)
    task.wait(1)
    timeLeft -= 1
end
```

```
-- buggy
while timeLeft > 0 do
  print(timeLeft)
  timeLeft -= 1
end
```

What is missing? What does the buggy countdown look like in the Output?

3 · Fill the Gap

The countdown should pause for 1 second between numbers. One word is missing. Fill it in using the word bank.

```
local timeLeft = 30

while timeLeft > 0 do
  print(timeLeft .. " seconds left!")
  task.____(1)
  timeLeft -= 1
end
```

Word bank: wait · stop · sleep

Write the missing word:

4 · Tell Me What You Built 📷

Now open your game and script your core mechanic using a pattern from the toolbox. When it works, come back here.

Send a photo or video of your mechanic working, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

My core mechanic is ...

I used the ... pattern ...

One tricky moment was when ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while your mechanic runs. Talk like you are teaching a friend. Try to use these words: **pattern**, **event**, **debounce**.

1. Play your game and show the core mechanic working.
2. Read your **Touched** line (or your loop) out loud.
3. Say which pattern you used from the toolbox.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.